

12

The Flame-Hand Letters of the Hebrew Alphabet

In the beginning of heaven and earth there were no words,
Words come out of the womb of matter.

Lao-Tse

12.1 Introduction

In Part One, I presented several examples of ways in which numbers and geometry may have been used by ancient civilizations. I also described Kepler's attempt to create a planetary system from the music of the spheres and Theodor Schwenk's and Lawrence Edwards' work to find unifying principles that generate natural forms. Projective geometry, harmonic law, systems of proportion, the Megalithic stone circles, and the designs found in the pavements of the Laurentian Library can be thought of as symbolic languages in their ability to express relationships found in art, architecture, music, design, and patterns in nature.

In Part Two I take a more formal approach to number and geometry and describe how certain systems evolve by continually feeding back information about themselves. I will show that the dynamics of such self-referential systems is governed once again by number and geometry. I ask the reader to ponder the question as to whether the patterns of number and geometry that we observe in the natural world are intrinsic to nature or do we subtly project aspects of our own thought processes onto the world?

The ultimate self-referential system is written language. The biologist, Lewis Thomas, has commented that written language functions much like a genetic code; it serves as a carrier of meaning and culture across the

generations. New meanings take the place of the old, yet remnants of the original meanings become part of the philological roots. In this chapter, I will present a hypothetical geometrical system developed proposed by Stan Tenen, an independent researcher, that attempts to shed light on the origin of the Hebrew alphabet and the meanings of its symbols [Ten1]. The letters are obtained by projecting a specially, meaningfully-shaped spiral, vortex form onto a flat surface from a variety of points of projection. The spiral vortex suggests a flame, and the projection source could be a flame, thus my reference to the letters as “flame letters”. Likewise, this vortex form can also be seen as an idealized human hand. When this model hand is placed on one’s hand, the wearer sees each letter when making particular gestures related to the meaning of the name of the letter, thus my reference to this model as a “flame-hand” model.

Tenen’s primary interest is in the recovery of what he takes to be an underlying geometric metaphor in Western traditions. My focus in this chapter is on the geometrical forms and mathematical content. Tenen’s story and mine share a common interest in the importance of systems created from within themselves, or self-referential systems. In the next chapter, I will study the mathematics of self-referential systems.

12.2 The Flame-Hand Letters of the Hebrew Alphabet

The Hebrew alphabet, or aleph–beth, plays an important role in the historical evolution of written language. We attribute the first fully symbolic written language to the Semitic tribes dating to 1500 B.CE. Before this time written language took the form of pictorial systems such as Egyptian hieroglyphics (which first appeared in 3000 B.CE.) in which stylized images of humans and human implements were interspersed with those of plants, birds, and other animals.

Although the Canaanite/Phoenician (“Old Hebrew”) written language is entirely symbolic, (every sound-syllable had its own character or letter), it does retain remnants of its pictorial predecessors. For example, aleph א (the equivalent of the English “A”) is the ancient word for ox (and the Canaanite Aleph looks like an ox-head on its side) while mem (the equivalent of “M”) is the Hebrew word for water and is symbolized by a

Table 12.1 Shadowgrams of the 27 Meruba Hebrew letters. They are all views of one physical system.

FLAME MODEL SHADOW- GRAMS	SAMPLE ALPHABETS FROM: "HEBREW ALPHABETS" c. 1950 REUBEN LEAF, BLOCH PUBLISHING				
	10-11th century Spanish Rabbinc 11 cent. "NACHMANIDES" "RASHI" Prototype	"RASHI"-style Typography	First printed Hebrew, Lisbon 1489	300 BCE, Elephantine Papyrus	
א	א	א	א	ALEF אֵלֶף	A
ב	ב	ב	ב	BET בֵּית	B
ג	ג	ג	ג	GIMEL גִּמְלָה	C
ד	ד	ד	ד	DALET דָּלֶת	D
ה	ה	ה	ה	HE הֵא	E
ו	ו	ו	ו	VAV וָו	F
ז	ז	ז	ז	ZAYIN זֵינַן	G
ח	ח	ח	ח	CHET חֵיט	H
ט	ט	ט	ט	TET טֵיט	J
י	י	י	י	YOD יֵוד	I
כ	כ	כ	כ	KAF כֵּף	K
ל	ל	ל	ל	LAMED לֵמֶד	L
מ	מ	מ	מ	MEM מֵם	M

Table 12.1 (Continued)

SAMPLE ALPHABETS FROM: "HEBREW ALPHABETS" c. 1950 REUBEN LEAF, BLOCH PUBLISHING					
FLAME MODEL SHADOW- GRAMS ↓	10-11th century Spanish Rabbinic 11 cent. "NACHMANIDES "RASHI" Prototype "RASHI"-style Typography First printed Hebrew, Lisbon 1489 300 BCE, Elephantine Papyrus				
	↓	↓	↓	↓	
נ	נ	נ	נ	נ	NUN נון N
ס	ס	ס	ס	ס	SAMEK סמך \$
ע	ע	ע	ע	ע	AYIN עין O
פ	פ	פ	פ	פ	PE פה P
צ	צ	צ	צ	צ	ZADI צדי Z
ק	ק	ק	ק	ק	QOF קוף Q
ר	ר	ר	ר	ר	RESH ריש R
ש	ש	ש	ש	ש	SHIN שין S
ת	ת	ת	ת	ת	TAV תו T
ך	ך	ך	ך	ך	KAF (FINAL) כף סוף U
ם	ם	ם	ם	ם	MEM (FINAL) מם סוף W
ן	ן	ן	ן	ן	NUN (FINAL) נן סוף X
ף	ף	ף	ף	ף	FE (FINAL PE) פה סוף V
ץ	ץ	ץ	ץ	ץ	ZADI (FINAL) צדי סוף Y

series of waves װ). Qof (the equivalent of “Q”) is the word for skull and is drawn with a circular top (the “skull”) and a vertical descender (the “spinal column”) ק.

It was only after the transfer of phonetic writing to Greece with the transformation of the aleph–beth into the Greek “alphabet”, that progressive abstraction of linguistic meaning reached completion. In his book, *The Spell of the Sensuous*, David Abram [Abr] traces the origins of written language and the effect this has had on the way we see the world.

The system proposed by Stan Tenen generates close facsimiles to the shapes of the 27 letters of the rabbinic form of the Meruba Hebrew alphabet shown in Table 12.1, while at the same time organizing them into a coherent pattern of meanings [Ten]. This system is also capable of generating the letters of particular forms of the Greek and Arabic alphabets. Tenen sees this as evidence of a common thread spanning many ancient systems of thought. The Hebrew letters in Table 12.1 are shown with English correspondences and correlations to the English alphabet (not all are phonetic equivalents).

It should be stated that Tenen’s proposal is historically controversial, and while it has begun to be presented for peer review, it would be premature to comment on its authenticity. There is little that scholars know for certain about the origins of Hebrew letters. Scholars have never been able to fully penetrate the meaning of particular sacred books of Jewish mysticism such as the Zohar, nor related ancient Sufi texts. Yet Tenen believes that these works become precise, unambiguous and meaningful when they are interpreted through their inherently geometrical metaphoric imagery.

It is unlikely that our forefathers understood the geometrical concepts that we shall present in the manner in which we understand them. After all, they had their own symbolic languages. However, mathematical concepts are fundamental and can be understood in many different ways. I shall present Tenen’s work as a kind of mathematical poem or metaphor. It will provide us with a model of a system generated from within itself.

12.3 The Vortex Defining the Living Fruit

Tenen first develops the concept of the life cycle of a living fruit. The idealized fruit is modeled by a “dimpled sphere” or *torus* (inner tube) whose

"Fruit tree yielding fruit whose seed is inside itself" Genesis 1, 11

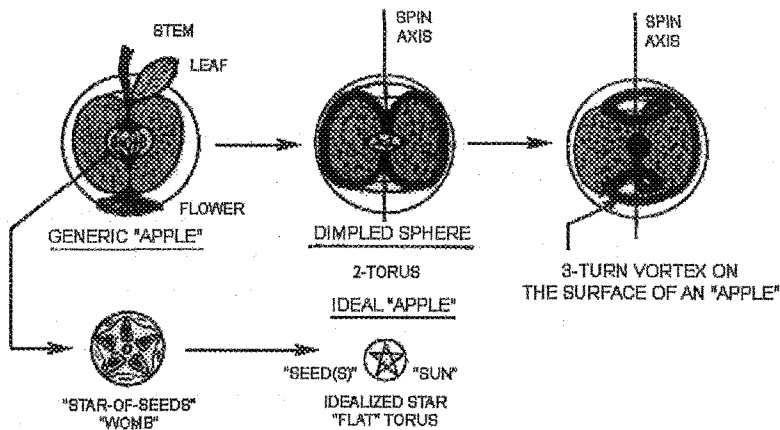


Figure 12.1 Idealized fruit modeled as a dimpled sphere.

inner radius shrinks to zero, as shown in Figure 12.1. The central sphere represents either the “seed-pod” of the current “fruit” or the next generation “fruit” nested within the current generation.

In vertical (stem up, flower down) cross section, the “dimpled sphere” or “generic apple” appears as two quasi-ellipses nested against and surrounding the central sphere, which appears as a circle sitting over the “kissing” point of the two ellipses. The ring of five seeds of the “apple” are arranged in a five-pointed star (the seeds in the core of an apple form a pentagon).

A seven-stage dynamic process through which the idealized fruit passes — the inherent propensity of the seed to grow — is modeled by considering the seed to be “containing” an internal increment (or quantum) of angular momentum. This is similar to the idea of *spin* in elementary particle physics, where the particle is considered to carry a quantum of spin in units of *angular momentum*. The seven-stage process unfolds as follows:

- 1) The potential angular momentum locked into the idealized seed-pod is expressed as an “impulse” that ejects the “sprout” from the “seed” as the

“fruit-tree” starts to grow. Initially all the angular momentum is expressed by the ejection of the “sprout” along its erupting “stem”.

- 2) Later in the life-cycle of the *fruit-tree*, when the *stem* or *trunk* of the tree has reached its maximum extent, the angular momentum — representing the life-force of the fruit-tree throughout its life-cycle — must be transferred into the unfurlment (around the tree-trunk) of the leaves and branches of the fruit-tree.
- 3) The branches sprout buds.
- 4) Which give rise to new fruit.
- 5) From which emerge flowers.
- 6) Still later, near the end of the life-cycle, the angular momentum in the volume of the mature fruit must be transferred to the next generation of seed (represented by the central sphere), as its *flowers* decay (and fall to the ground, now devoid of *life-force* and angular momentum), and as this new seed (containing the passed-on life-force and increment of angular momentum of the life-cycle) is thrown to the wind.
- 7) The new seed falls to the ground to restart the next life-cycle.

The Sefer Zohar, quoting Genesis, describes the entire living system as a “fruit-tree, yielding fruit, whose seed is in itself”. Each generation of fruit, in turn, is seen as nested within the previous generation giving rising to the recursive sequence: ...-acorn-oak-acorn-oak-.... We have here the quintessential self-referential process in which fruit is both a vessel and its contents. (The vessel is modeled as a torus, which as with vortices (see Section 1.11), is a minimal closed system, the first requirement for self-reference.) For example, a smoke ring maintains its integrity in the form of a torus.

The process of unfurlment and spin shown in Figure 12.2 is modeled by a ribbon with three turns. Since the original seed and the new seed within the new whole fruit are identical in their development, except for being a generation apart, they can be taken to be the ends of the developmental life-cycle strung between them. The ribbon represents the exchange and transformation of angular momentum that takes place during the seven-stage life cycle. This vortex form can also be taken to represent an individual. The living spirit of the individual is symbolized by his/her whirling motion as in Sufi or Dervish dancing. The philosopher, mathematician,

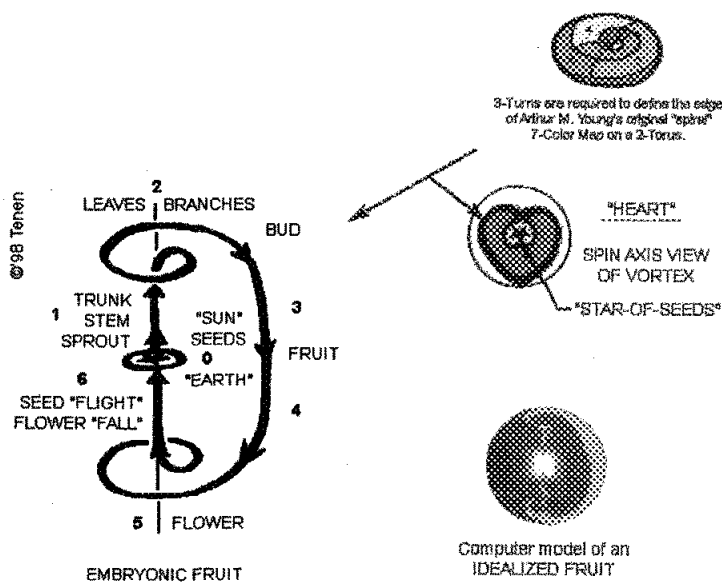


Figure 12.2 Cycle of life of a living fruit. The idealized embryology of a fruit-tree is mapped onto an ideal fruit.'

and engineer Arthur Young [Young] has written extensively in his book, *The Reflexive Universe*, about the seven-stage process under which all natural forms of the universe unfold. Tenen credits this system, and his fruit tree conforms well to it.

12.4 The Torus

The *torus* is the first structure of mathematical importance that we encounter in Tenen's proposal. What is a torus? Any circular (loop) cut made on the surface of a sphere divides the sphere into two pieces. However, Figure 12.3 shows that two loop cuts leave the surface of a torus (inner tube) in one piece and opens the torus up to a *period rectangle*. Each side of the period rectangle is *identified* with its opposite, since they represent the two loop cuts. The two circles that characterize the torus are its *meridian* and *longitude*, shown in Figure 12.4.

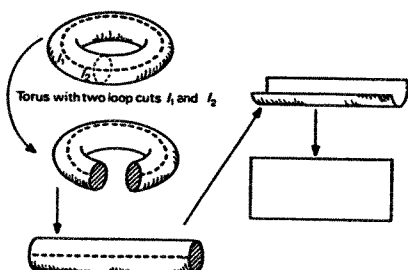


Figure 12.3 A torus is opened to a period rectangle by cutting two loops on its surface.

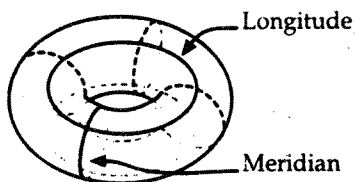


Figure 12.4 A meridian and longitude on a torus.

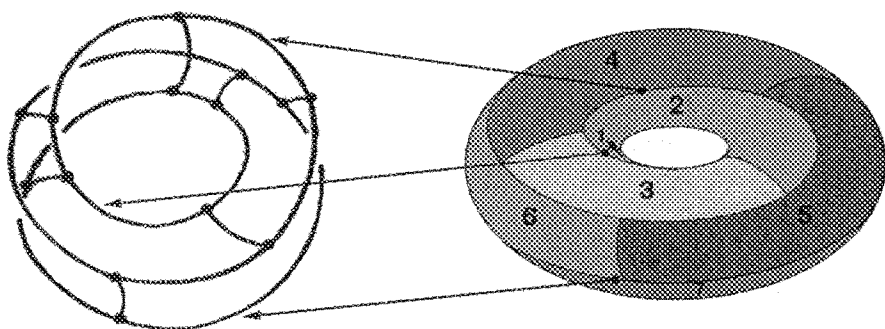
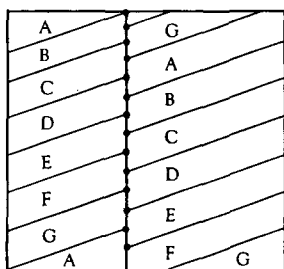


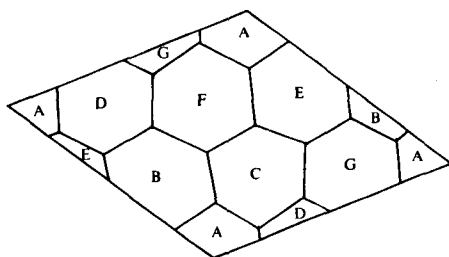
Figure 12.5 The surface of a torus divides itself naturally into seven regions.

In Figure 12.5, due to Arthur Young, we see that the ribbon of Figure 12.2 not only defines the structure of the torus but divides its surface in a natural way into seven regions. This is the *seven-color map* of the torus which states that a torus can be subdivided into seven regions, each region sharing an edge with the other six. If such a map were to be colored so that regions sharing an edge have different colors, clearly seven colors are needed. (It is well known that any map drawn on a sphere or the plane never requires more than four colors.)

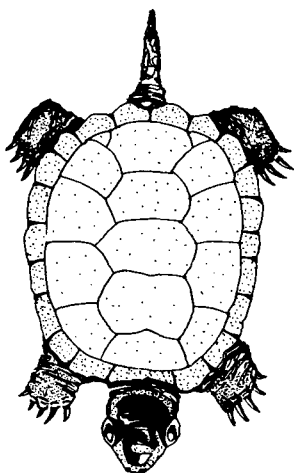
Figure 12.6a shows a seven-color map redrawn on a period rectangle. Notice how the seven regions continue from the left to the right edge, and



(a)



(b)



(c)

Figure 12.6 (a) and (b) seven-color map drawn as hexagons on a period rectangle; (c) hexagonal design on the shell of a turtle.

from the top to the bottom edge. Also notice that they are all hexagons (see Figure 12.6b). Tenen believes that this association of the torus with hexagons may lie at the basis of the ancient myths in which the world is brought into being on the back of a turtle whose shell is made of hexagons (see Figure 12.6c).

12.5 The Tetrahelix

A *tetrahelix* is a spiral column formed by combining a column of tetrahedra joined face to face, as shown in Figure 12.7. Three continuous spiraling ribs can be seen on the surface of the tetrahelix, forming a double helix. The tetrahelix can also be imagined to be formed by taking a triangular prism (see Figure 12.8) and giving it a $\frac{1}{3}$ -turn so that top face is rotated 120 degrees with respect to the bottom face for each 11-tetrahedra. Actually, no integral number of tetrahedra results in a top face oriented identical to

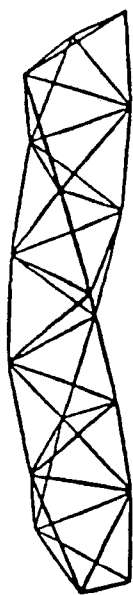


Figure 12.7 A tetrahelix column.

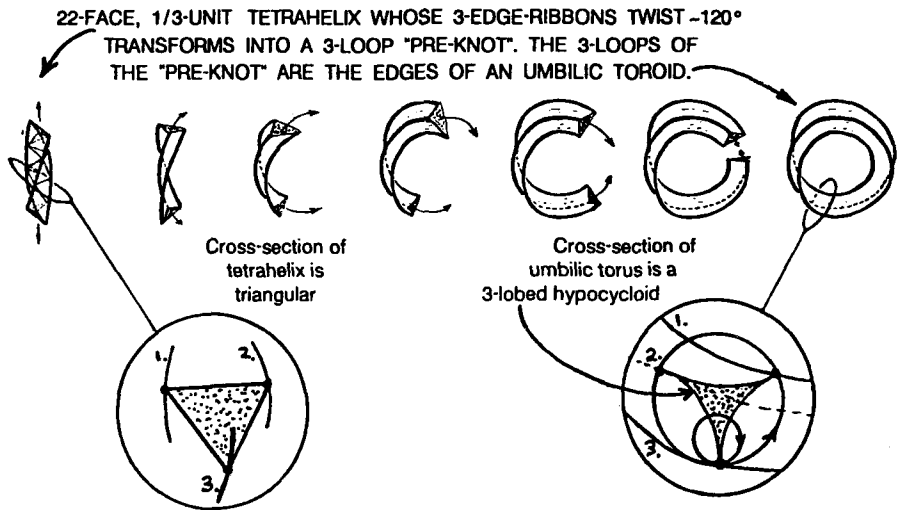


Figure 12.8 A triangular prism is twisted into a tetrahelix and then rotated to form an umbilic toroid. Thirty three tetrahedra result in a column in which top and bottom faces are rotated approximately 360 degree.

the bottom face. In other words, the tetrahelix is not a *periodic* structure and so the column must be slightly strained to enable the top and bottom faces to meet. A tetrahelix with 33 tetrahedra results in a column in which top and bottom faces are rotated approximately 360 degrees with respect to each other. Tenen has shown that the two ends of the $\frac{1}{3}$ -unit tetrahelix can then be bent around in a circular arc and joined together to form an "umbilic torus". Now the three disjoint spirals on the tetrahelix column form one continuous arc with three loops on the umbilic torus that precisely defines the edges of the seven color map on the torus (see Figure 12.5). A tetrahelix with 11-tetrahedra can be easily constructed by folding the triangular grid shown in Figure 12.9 up from the plane.

Notice that the outline of this folding pattern is a hexagon which can also tile the plane, as shown in Figure 12.10a (see also Figure 12.6). Notice how the $\frac{1}{3}$ -unit of flattened tetrahelices form the six-around-one pattern of the seven-color map that characterizes the surface of a torus. Figure 12.10b shows seven clusters of seven clusters of seven $\frac{1}{3}$ -unit tetrahelices.

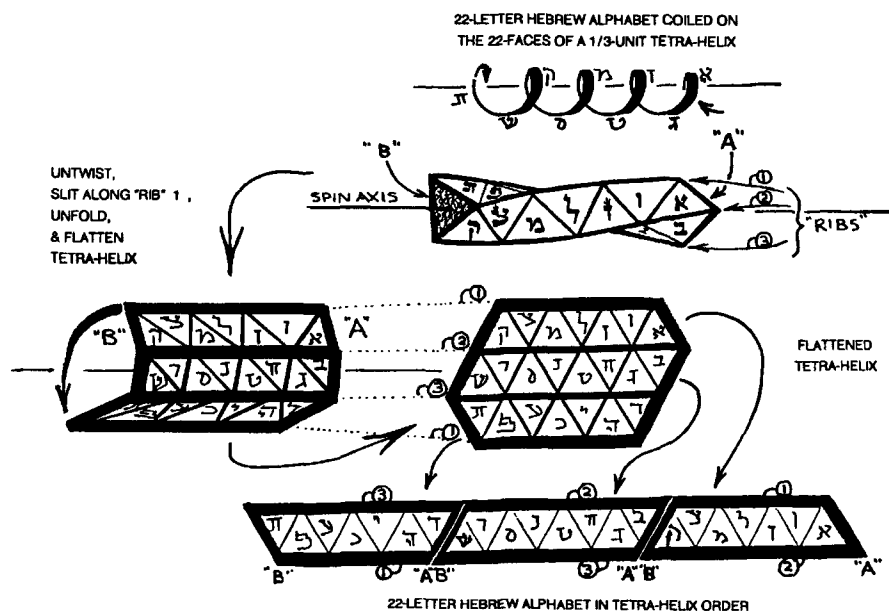
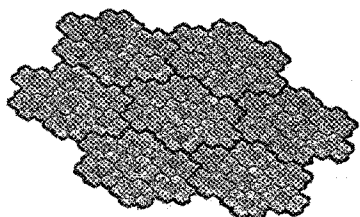


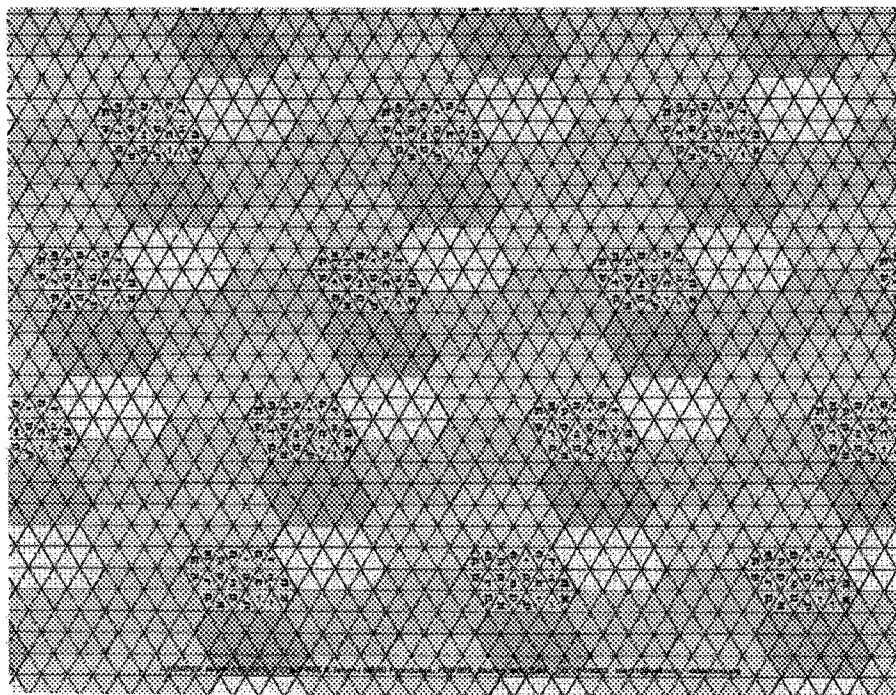
Figure 12.9 Paper folding construction of a tetrahelix with 11 tetrahedra.

We could make groups of seven at higher and higher levels indefinitely. The curve along the edge of this figure is called a “flowsnake” and is named after the “snowflake”, or Koch curve, to which it is related (see Section 18.5). The 22 letters of the Hebrew alphabet “seed” this hierarchy.

When bent around and connected, the 11 tetrahedra exhibit 22 external faces (see Figure 12.9). Tenen has placed the sequence of 22 letters of the Hebrew alphabet (not counting the final letter forms) along these faces in alphabetical order. Notice that the letters of the Hebrew alphabet occur in sequential order on the hexagonal tiling of Figure 12.10b. In Figure 12.11, the edge of the umbilic torus has been isolated as a string, and Tenen has strung all 27 sletters of the Hebrew alphabet on the three turns of the string with nine letters for each of the three loops. In Figure 12.12 the second (bet) and last letter (tav) are merged to form the first letter (aleph) so that aleph is seen as both “head” and “tail” of the “Oroboros” or “snake that eats its tail”.



(a)



(b)

Figure 12.10 (a) Seven clusters of seven tetrahelices forming a fractal curve called the "flowsnake"; (b) the Hebrew letters superimposed on the hexagonal tiling.

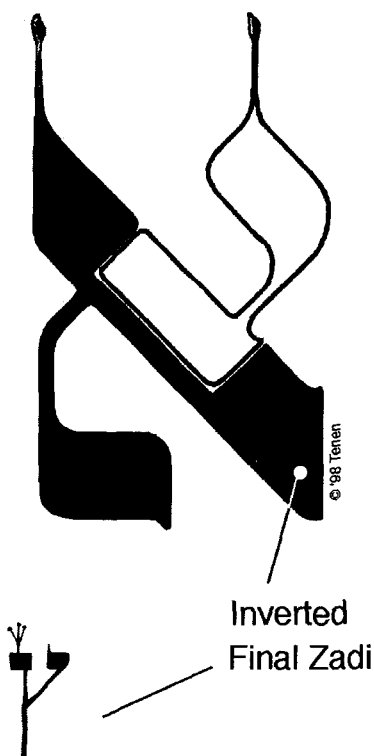


Figure 12.12 The second letter of the Hebrew alphabet (bet) combines with the last letter (tav) to form the first letter (aleph).

12.6 The Meaning of the Letters

The umbilic torus necklace divides the alphabet naturally into three parts, to form a connected *enneagon* (circular nine-pointed figure). Each triple of letters in the enneagon can be associated with a common meaning but on three different levels. The outer level, or level 1, is the archetypal meaning; level 2 is inner/spiritual, while level 3 is outer/physical. Also notice in Figure 12.11 the left-right symmetry at each level. Tenen sees this process of moving from “oneness” to “wholeness” as represented by the 27 letters of the Hebrew alphabet (including final forms);

Oneness — Aleph, Yod, Qof;

Distinction — Bet, Kaf, Resh;

Action — Gimel, Lamed, Shin;

Division — Dalet, Mem, Tof;

Connection — Heh, Nun, Kaf-final;

Multiplication — Vav, Samek, Mem-final;

Projection — Zayin, Ayin, Nun-final;

Encompassment — Chet, Peh, Peh-final;

Wholeness — Tet, Zadi, Zadi-final.

So we have here the makings of a sacred alphabet, in which the letters are not merely abstract symbolic forms but also have meaning in and of themselves. Let's look in detail at the meaning of the first two letters of the alphabet at levels 1, 2, and 3.

12.6.1 *Oneness*

Level 1. Aleph Absolute, stands for All, Aloof, and Alp — a high point, a high mountain. It is All-One-Wisdom-Consciousness. Phela (Aleph in reverse) is a miracle and a mystery. Aleph as the singular is the sun-seed-center (the primal point), the Source; Aleph as the Whole, is the idealized Fruit — the Apple. Aleph represents the strongest most coherent Archetype. That is why Aleph means Ox or Master.

Level 2. Yod Our personal, individual consciousness is represented by the human hand, which projects our personal consciousness into the physical world. The Yod is the “seed” and is therefore associated with the male organ and with semen and it is a point (iota in Greek). Yod is an expression of our being (the psychologist's Id). That is why Yod means Consciousness, hand or point/pointer.

Level 3. Qof This is our outer “mechanical” or Monkey consciousness. It is the shell or physical copy of our inner Yod consciousness. When we “ape”, we Copy. That is why Qof means Monkey, Copy or Skull.

Together Qia (Aleph-Yod-Qof) means Eruption. Qi is the name for the life-force in the Eastern traditions and “eruption” is what happens at the seed-center of Continuous creation — at the center of the torus.

12.6.2 Distinction

Level 1. Bet Partitions the Unity and Wholeness of Aleph. Aleph by itself is One. There is no opposition or polarity, and no need for action or change. Bet as the first possible distinction differentiates what is within from what is without. In his *Laws of Form*, G. Spencer-Brown [Spe-B] showed that a single mark of distinction separating inside from outside was capable of reproducing all of the laws of logic. From Bet on there is difference, complement, and contrast. There is spirit and matter, wave and particle. That is why Bet, two and duality is a “Housing” (which separates inside from outside). As a prefix, Bet means In or With.

Level 2. Kaf As the inner aspect of Bet, represents holding in, as in cupping in the palm of the hand. Cupping shapes the palm like that which it holds. That is why Kaf means palm and why, as a prefix, it designates Likeness and Similarity.

Level 3. Resh As the outer part of Bet, Resh represents the outer reaching of Bet. If Kaf is what is in the palm of the hand, then Resh is what radiates from the head. That is why Resh means Head, Reaching Rushing and Radiation. Together Buker (Bet–Kaf–Resh) means “first born son”. These letters break open Unity and signify the birth of distinction at each of their levels.

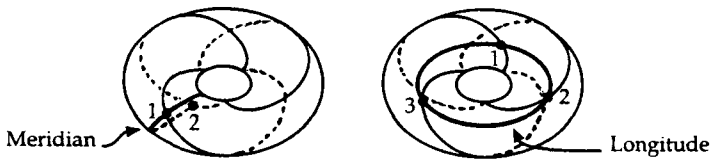
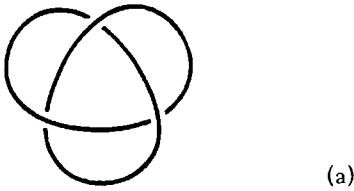
In a similar manner Tenen has shown that all of the letters of the Hebrew alphabet participate in this three level process that integrates the inner world of our consciousness with the outer physical world. Mastery of the inner world leads to wisdom, while mastery of the outer world leads to understanding.

12.7 Generation of the Flame-Hand Letters

Tenen proposes that the rabbinic form of the Meruba Hebrew letters can be generated from a *knot* drawn on a dimpled-sphere-shaped *torus*. In order to better understand this construction, let us first consider the concept of a knot.



Figure 12.13 (a) Example of a trivial knot or “unknot”; (b) the unknot drawn with a single crossover.



The trefoil is a (3, 2)-torus knot.

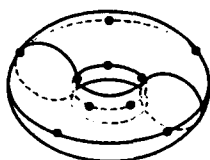
(b)

Figure 12.14 (a) The trefoil knot, the simplest nontrivial knot, drawn as an under-over-under pattern; (b) the trefoil knot on the surface of a torus.

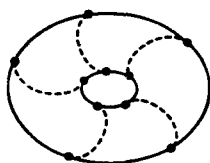
The simplest form of a knot is gotten by connecting the ends of a string together to form a loop or *unknot* (see Figure 12.13a) [AdamC]. In Figure 12.13b the unknot, or trivial knot as it is called, is redrawn with a single crossing. Since the crossing can be removed without cutting the string, it is considered to be structurally identical or isomorphic to the unknot. The edge of the umbilic torus is an unknot.

The first nontrivial knot is the *trefoil knot* shown in Figure 12.14a with three crossings, none of which can be removed. The trefoil knot, as with

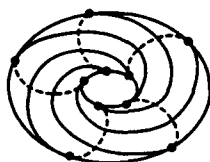
all nontrivial knots, is basically three-dimensional. However, to represent it schematically, mathematicians generally draw its projection in two-dimensional space as an under-over-under pattern in which a broken line indicates that the string passes under the over-stretched arc (see Figure 12.14b). The trefoil map is redrawn around the surface of a torus in Figure 12.14c. Notice that in a torus knot there are no crossovers. The trefoil torus knot is called $(3, 2)$, since every longitude of the torus intersects the knot three times while every meridian intersects it twice. A $(5, 3)$ knot is shown in Figure 12.15. To draw this knot, place 5 evenly-spaced vertices around the inner circle of the torus and 5 vertices on the outer circle, directly opposite them. Connect any vertex on the inner circle with the vertex on the outer circle displaced by 3 from it in a counterclockwise direction.



Marking points on the equators.



Attach points by strands across the bottom of the torus.



Constructing a $(5, 3)$ -torus knot.

Figure 12.15 Steps to drawing a $(5, 3)$ knot on the torus.

A torus knot can also be generated from a tetrahelix column by bending a column formed of multiples of 11 tetrahedra around to a torus, as we did for the umbilic torus. The spirals on the surface of the tetrahelix form continuous curves on the surface of the torus in the form of $(n, 3)$ torus

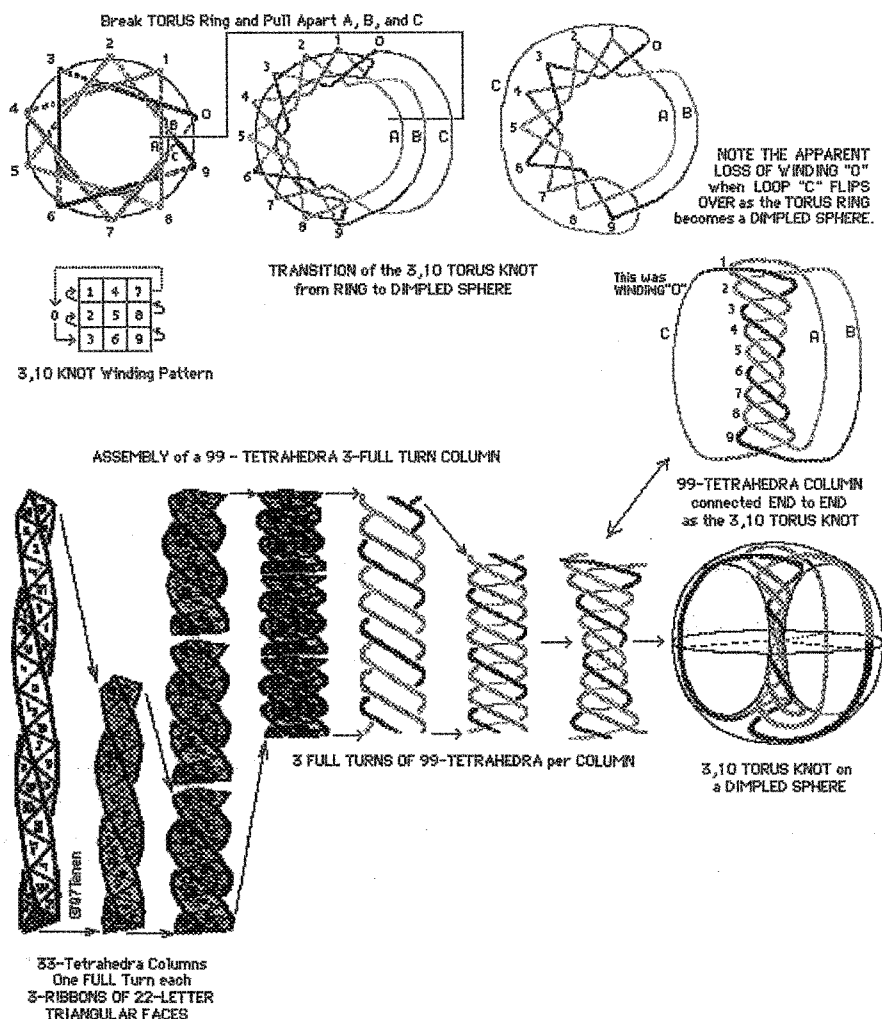


Figure 12.16 Transition from a $(10, 3)$ and a 33-tetrahedra column to a dimpled sphere in the form of a braided structure.

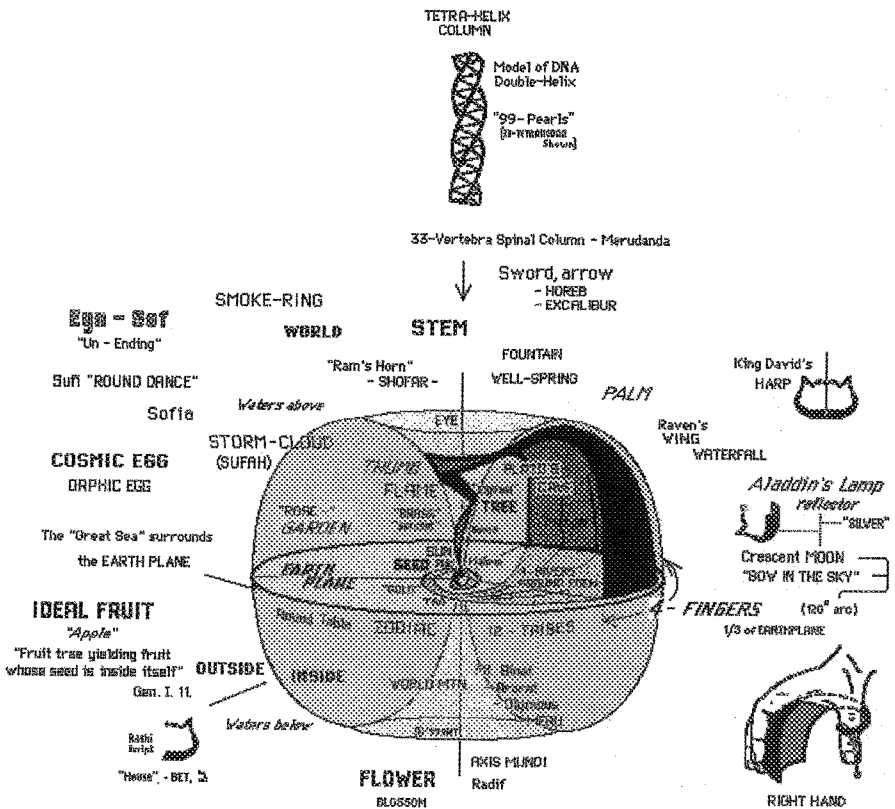


Figure 12.17 Creation of a hand-like structure emanating from the seed of the “fruit tree”.

knots for n , an integer greater than 2 and not divisible by 3, where $n-1$ is the number of multiples of 11 tetrahedra that form the tetrahelix when $n > 3$ (the tetrahelix makes more than one complete turn). Two special cases are the tri-loop (1, 3) or umbilic torus formed from 11 tetrahedra, and the trefoil knot (2, 3) formed from 22 tetrahedra.

The transition from a $(10, 3)$ knot to a dimpled sphere in the form of a braided structure with three entwined strands is shown in Figure 12.16. The three strands of this braided structure span six hand-like structures that encompass the generic apple shown in Figure 12.17. Three hands around the top and three around the bottom of the apple surround the seed at the



Figure 12.18 Image of a hypersphere.

center. From one of the strands, Tenen creates the hand-like structure shown emanating from the “seed” of the “fruit tree” with a protruding spiral “thumb”. It is beyond the scope of this book to describe the spiral component of the hand in detail. However, its projection onto the plane can be expressed when the spiral is properly truncated, by the simple equation $r = \frac{1}{\theta}$ in polar coordinates. This is the same spiral curve illustrated in Figure 4.6 as part of the “eye of Horus”. The *reciprocal or hyperbolic* spiral transitions smoothly from being asymptotic to a line, to being asymptotic to a circle. Unlike the common logarithmic spirals, the reciprocal spiral is completely asymmetrical — and this is essential to Tenen’s proposal. Tenen also suggests that his “generic apple” or “dimpled sphere” is actually intended to represent the three dimensional projection of the surface of a four-dimensional sphere or hypersphere an image of which is shown in Figure 12.18. (This is also presented by Arthur M. Young in *The Reflexive Universe*.)

This vortex-like structure forms the “flame-hand” from which Tenen projects all of the letters of the Hebrew alphabet. To get some idea as to the versatility of this asymmetric form, Tenen places it within a tetrahedron, the simplest embodiment of symmetry (see Figure 12.19). He likens this metaphorically to the Old Testament reference to the “light in the meeting tent” in which the light is sometimes referred to as a “flame”, while the tent is referred to as a “coal”. A projected image of the flame is taken in the direction of the tetrahedron’s seven axes of rotational symmetry, giving rise to the seven distinctly different images shown in Figure 12.20.

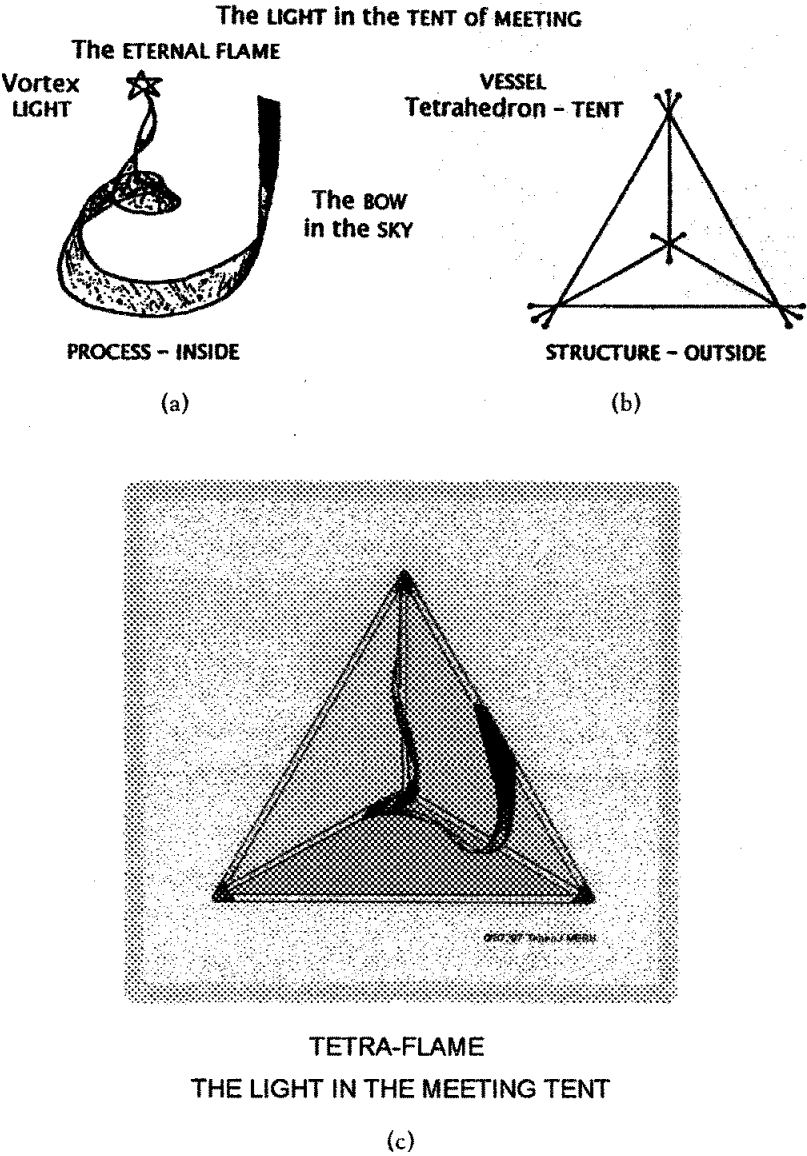


Figure 12.19 (a) The hand model, or vortex, is referred to as the “flame”; (b) the tetrahedron is the “tent”; (c) the combination represents the “light in the meeting tent” from Exodus.

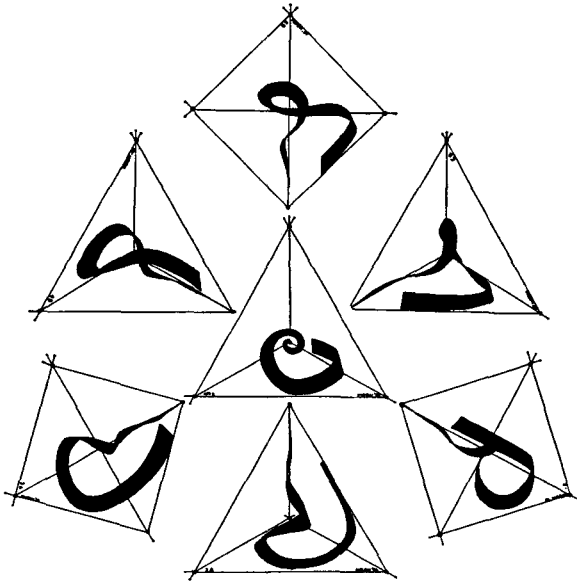


Figure 12.20 Flame images projected in the direction of the seven axes of a tetrahedron.

According to some scholars, up until the Babylonian exile, the Old Hebrew alphabet — that is the proto-Siniatic/Canaanite alphabet — consisted of 22 pictograms. During the Babylonian exile, about 500 B.C.E. a new *square-form* alphabet came into use. The Assyrian-Babylonian square-form Meruba letters replaced the Old Hebrew letters in sacred usage. The new alphabet now had 27 letters, in which five additional final-form letters were added to the original 22. Early examples of the Meruba letters were found at Elephantine (circa 300 B.C.E.). These letters were also very similar to examples of the Rashi-Nachmanides style of rabbinic handwriting in use in Islamic Spain, but somewhat different from the formalized “squared-off” forms of the Hebrew letters used on modern Torah scrolls to reduce the possibility of ambiguous readings. The letters in the first column of Table 12.1 were created from projections of Tenen’s hand form. Notice the similarity to the Spanish-era rabbinic forms of the letters, and even the somewhat distant relationship to the Elephantine script.

Tenen has gone another step with his proposal. He has found that the names of the letters give information as to a series of hand positions in

which the eye projects his letter generator to the form of that letter. For example when the position of the hand is in the area of the mouth, its projection creates the letter “peh” (PAY) which means mouth. He has also used his models to discover letter-level coding in the Hebrew text of Genesis in the Bible, and he feels that the sequence of letters in the Torah may have served, through his hand gestures, as a kind of sacred meditation. He feels that it is the hand that projects consciousness from the inner to the outer worlds. However these ideas take us beyond the scope of this book and my own personal knowledge base.

12.8 Some Commentary on Tenen’s Proposal

As one would do for poetry, Tenen’s proposal of flame-hand letters should be evaluated not only on its literal meanings but on its ability to ring true at the level of metaphor. How well does it explain sacred texts and how does it speak to our desire to make sense out of our present world?

Although Tenen has found no explicit evidence for the use of the hand-gesture system for generating the letters, he has found references that point to the use of the reciprocal spiral (circle into line) in forming the letters, and he has found allusions to this in the so-called “Credo” of Judaism, the Sh’ma, from Deuteronomy, and elsewhere in ancient texts and practices. It is well known that Orthodox Jews wear two small cubical leather boxes with passages from the Hebrew Bible on their arms and foreheads during morning prayers. This is described immediately after the opening lines of the Sh’ma, and the descriptive text is what is placed in the two boxes.

The arm-*tefillin* (box with scripture) is held onto the upper arm by a leather strap, which is wound on the arm seven times, and then wrapped on the hand to form the outline shapes of some Hebrew letters, usually Shin, Dalet, and Yud, which spell the Hebrew word “Almighty” (a God-name). The letter Shin also appears on the tefillin-box worn on the forehead, and additional letters are sometimes found in the knot on the strap holding the tefillin-box on the forehead. So, in effect, Tenen is proposing an underlying source for the tefillin-hand-alphabet tradition, which is retained in the way tefillin are used by observant Jews today.

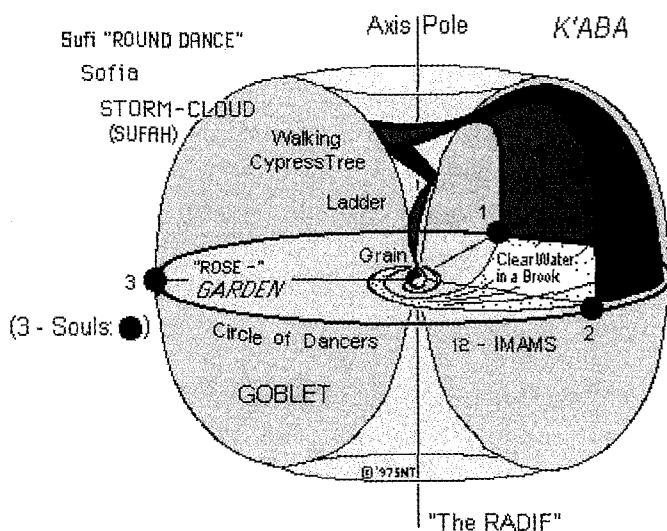


Figure 12.21 Tenen's hand figure captures the imagery of Rumi's verse.

There is also much imagery in the poetry of the medieval Islamic poet Rumi that could refer to a structure akin to Tenen's hand model. Some excerpts of Rumi's verse follow, correlated with the geometrical images in Figure 12.21:

Come, Come! Let us whirl about in the rose-garden. —

Let us now whirl about the grain
which no granary comprehends. —

I am like a goblet
in the circle of dancers,
turning from one hand to the other
with my story. —

Come, come, O thou, who are the soul
of the soul of the soul of the round dance.

Come thou who are the walking cypress
in the garden of the round dance.

Come thou, under whose feet is the fountain of light.

The roof of the seventh sky —

does not reach where reaches the ladder of the round-dance.

12.9 Conclusion

Tenen's metaphor relating the genesis of the Hebrew letters to the creation story relates, at least on the surface, to modern theories of consciousness and self-referential systems. In the next chapter knots will be used to mathematically characterize self-referential systems. Similar mathematical structures to the ones that Tenen has chosen to use for his proposal arise in other systems related to recent research into the nature of consciousness that is the subject of the next chapter.

Tenen's suggestion that the ancients made use of a torus knot ("basket-weaving") is also suggestive of modern theories of plant growth or phyllotaxis that we shall present in Chapter 24. The human heart is also toroidal, wound with microscopic tubular muscles [Pes].